INT202 Complexity of Algorithms

1. Use Euclidean Algorithm to find GCD of 134 and 52

Solution:

r2 r1 q r

$$134 = 52 \cdot 2 + 30$$

$$52 = 30 \cdot 1 + 22$$

$$30 = 22 \cdot 1 + 8$$

$$22 = 8 \cdot 2 + 6$$

$$8 = 6 \cdot 1 + \boxed{2} \longrightarrow \text{GCD}$$

$$6 = 2 \cdot 3$$

2. Given two integers 134 and 52, find two integers, s and t, such that s*a+t*b=gcd(a,b)

q	r1	r2	r	s1	s2	s	t1	t2	t
2	134	52	30	1	0	1	0	1	-2
1	52	30	22	0	1	-1	1	-2	3
1	30	22	8	1	-1	2	-2	3	-5
2	22	8	6	-1	2	-5	3	-5	13
1	8	6	2	2	-5	7	-5	13	-18
3	6	2	0	-5	7	-26	13	-18	67
	2			7			-18		

2 = 7 · 134 + 52 · (-18).

3. Find the multiplicative inverse of 8 mod 11.

q	r1	r2	r	t1	t2	t
1	11	8	3	0	1	-1
2	8	3	2	1	-1	3
1	3	2	1	-1	3	-4
2	2	1	0	3	-4	11
	1			-4		

8*(-4) mod 11 ≡ 1 mod 11.

-4 mod 11 ≡ 7 mod 11.

So the multiplicative inverse 8 mod 11 is 7.

4. Let n > 0 be an integer. Prove that n is divisible by 9 if and only if the sum of its digits is divisible by 9.

Solution:

1. **Decimal Expansion:**

   The positive integer n is expressed in base-10 as:

$$n = \sum_{k=0}^{m} a_k 10^k, \quad a_k \in \{0, 1, \ldots, 9\}.$$

2. **Modulo 9 Property:**

   The key observation is $10 \equiv 1 \pmod 9$, which implies for any $k \geq 0$:

$$10^k \equiv 1^k \equiv 1 \pmod 9.$$

3. **Derivation of $n \mod 9$:**

   Taking n modulo 9:

$$n \equiv \sum_{k=0}^{m} a_k 10^k \equiv \sum_{k=0}^{m} a_k \cdot 1 \equiv \sum_{k=0}^{m} a_k \pmod 9.$$

4. **Divisibility Condition:**
   - $n \equiv 0 \pmod 9 \iff \sum_{k=0}^{m} a_k \equiv 0 \pmod 9$.
   - That is, $9 \mid n \iff 9 \mid (\text{sum of digits})$.

5. Solve the equation: **$4x + 7 \equiv 6$ (mod 11).**

Solution:

Add -7 to both sides: $4x \equiv -1 \pmod{11}$

gcd(4, 11)=1, there is only one solution

$x = (-1\times 4^{-1}) \bmod 11 = (-1\times 3) \bmod 11 = 8$

Verify the solution: $4\times 8+7= 39$, $39-11\times 3=6$. so $4\times 8+7\equiv 6 \pmod{11}$.